

TUGAS PEKAN 7 STRUKDAT

IMPLEMENTASI SORTING ALGORITHM (SORTING DATA MAHASISWA)

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1. KODE PROGRAM 1 (CLASS METHOD)

```
2. package pekan7_2511533024;
3.
4. public class Mahasiswa_2511533024 {
5.     private String name_3024;
6.     private String nim_3024;
7.     private String prodi_3024;
8.
9.     public Mahasiswa_2511533024(String name_3024, String nim_3024, String
prodi_3024) {
10.         this.name_3024 = name_3024;
11.         this.nim_3024 = nim_3024;
12.         this.prodi_3024 = prodi_3024;
13.     }
14.
15.     public String getName_3024() {
16.         return name_3024;
17.     }
18.
19.     public void setName_3024(String name_3024) {
20.         this.name_3024 = name_3024;
21.     }
22.
23.     public String getNim_3024() {
24.         return nim_3024;
25.     }
26.
27.     public void setNim_3024(String nim_3024) {
28.         this.nim_3024 = nim_3024;
29.     }
30.
31.     public String getProdi_3024() {
32.         return prodi_3024;
33.     }
34.
35.     public void setProdi_3024(String prodi_3024) {
36.         this.prodi_3024 = prodi_3024;
37.     }
38.
39.     @Override
40.     public String toString() {
41.         return name_3024 + "(" + nim_3024 + ")";
42.     }
43. }
```

2. KODE PROGRAM 2 (CLASS MAIN)

```
1. package pekan7_2511533024;
2.
3. import javax.swing.*;
4. import javax.swing.table.*;
5. import java.awt.*;
6. import java.util.*;
7.
8. public class SortingProgram_2511533024 extends JFrame {
9.
10.     private JTextField txtName_3024, txtNim_3024, txtProdi_3024;
11.     private JButton btnAdd_3024, btnDelete_3024, btnSort_3024;
12.     private JComboBox<String> cmbAlgo_3024;
13.     private JTable tableMahasiswa_3024;
14.     private DefaultTableModel tableModel_3024;
15.     private JTextArea taVisualisasi_3024;
16.     private ArrayList<Mahasiswa_2511533024> dataList_3024;
17.
18.     public SortingProgram_2511533024() {
19.         dataList_3024 = new ArrayList<>();
20.         initComponents_3024();
21.         loadDefaultData_3024();
22.         refreshTable_3024();
23.     }
24.
25.     private void initComponents_3024() {
26.         setTitle("Sorting Mahasiswa - 2511533024");
27.         setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
28.         setLayout(new BorderLayout());
29.
30.         JPanel panelInput_3024 = new JPanel(new GridLayout(4, 2, 5, 5));
31.         panelInput_3024.setBorder(BorderFactory.createTitledBorder("Input
Data Mahasiswa"));
32.
33.         panelInput_3024.add(new JLabel("Nama:"));
34.         txtName_3024 = new JTextField();
35.         panelInput_3024.add(txtName_3024);
36.
37.         panelInput_3024.add(new JLabel("NIM:"));
38.         txtNim_3024 = new JTextField();
39.         panelInput_3024.add(txtNim_3024);
40.
41.         panelInput_3024.add(new JLabel("Program Studi:"));
42.         txtProdi_3024 = new JTextField();
43.         panelInput_3024.add(txtProdi_3024);
44.
45.         panelInput_3024.add(new JLabel(""));
46.         btnAdd_3024 = new JButton("Tambah");
47.         panelInput_3024.add(btnAdd_3024);
48.
49.         JPanel panelSort_3024 = new JPanel(new FlowLayout());
50.         panelSort_3024.setBorder(BorderFactory.createTitledBorder("Pengurutan"));
```

```

51.
52.     panelSort_3024.add(new JLabel("Pilih Algoritma:"));
53.     String[] algo_3024 = {"Insertion Sort", "Selection Sort", "Bubble
Sort"};
54.     cmbAlgo_3024 = new JComboBox<>(algo_3024);
55.     panelSort_3024.add(cmbAlgo_3024);
56.
57.     btnSort_3024 = new JButton("Mulai Sorting");
58.     btnDelete_3024 = new JButton("Hapus Terpilih");
59.
60.     panelSort_3024.add(btnSort_3024);
61.     panelSort_3024.add(btnDelete_3024);
62.
63.     tableModel_3024 = new DefaultTableModel(new String[]{"Nama", "NIM",
"Program Studi"}, 0);
64.     tableMahasiswa_3024 = new JTable(tableModel_3024);
65.
66.     JScrollPane scrollTable_3024 = new
JScrollPane(tableMahasiswa_3024);
67.     scrollTable_3024.setBorder(BorderFactory.createTitledBorder("Daftar
Mahasiswa"));
68.
69.     taVisualisasi_3024 = new JTextArea();
70.     taVisualisasi_3024.setEditable(false);
71.     taVisualisasi_3024.setFont(new Font("Monospaced", Font.PLAIN, 12));
72.     JScrollPane scrollVisual_3024 = new
JScrollPane(taVisualisasi_3024);
73.     scrollVisual_3024.setBorder(BorderFactory.createTitledBorder("Proses
Sorting (Langkah-langkah)"));
74.
75.     JPanel panelTop_3024 = new JPanel(new BorderLayout());
76.     panelTop_3024.add(panelInput_3024, BorderLayout.NORTH);
77.     panelTop_3024.add(panelSort_3024, BorderLayout.SOUTH);
78.
79.     JSplitPane splitPane_3024 = new
JSplitPane(JSplitPane.VERTICAL_SPLIT, scrollTable_3024, scrollVisual_3024);
80.     splitPane_3024.setResizeWeight(0.5);
81.
82.     add(panelTop_3024, BorderLayout.NORTH);
83.     add(splitPane_3024, BorderLayout.CENTER);
84.
85.     btnAdd_3024.addActionListener(e -> btnAddActionPerformed_3024());
86.     btnDelete_3024.addActionListener(e ->
btnDeleteActionPerformed_3024());
87.     btnSort_3024.addActionListener(e -> btnSortActionPerformed_3024());
88.
89.     setSize(900, 600);
90.     setLocationRelativeTo(null);
91. }
92.
93. private void loadDefaultData_3024() {
94.     dataList_3024.add(new Mahasiswa_2511533024("Budi", "001",
"Informatika"));

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95.         dataList_3024.add(new Mahasiswa_2511533024("rizky", "002", "Sistem
Informasi"));
96.         dataList_3024.add(new Mahasiswa_2511533024("Maya", "003", "Ilmu
Komputer"));
97.         dataList_3024.add(new Mahasiswa_2511533024("andi", "004", "Teknik
Elektro"));
98.         dataList_3024.add(new Mahasiswa_2511533024("Citra", "005",
"Manajemen Informatika"));
99.     }
100.
101.     private void refreshTable_3024() {
102.         tableModel_3024.setRowCount(0);
103.         for (Mahasiswa_2511533024 m_3024 : dataList_3024) {
104.             tableModel_3024.addRow(new Object[]{
105.                 m_3024.getName_3024(),
106.                 m_3024.getNim_3024(),
107.                 m_3024.getProdi_3024()
108.             });
109.         }
110.     }
111.
112.     private void btnAddActionPerformed_3024() {
113.         String name_3024 = txtName_3024.getText().trim();
114.         String nim_3024 = txtNim_3024.getText().trim();
115.         String prodi_3024 = txtProdi_3024.getText().trim();
116.
117.         if (name_3024.isEmpty() || nim_3024.isEmpty() ||
prodi_3024.isEmpty()) {
118.             JOptionPane.showMessageDialog(this, "Semua field harus
diisi!", "Error", JOptionPane.ERROR_MESSAGE);
119.             return;
120.         }
121.
122.         dataList_3024.add(new Mahasiswa_2511533024(name_3024,
nim_3024, prodi_3024));
123.         refreshTable_3024();
124.
125.         txtName_3024.setText("");
126.         txtNim_3024.setText("");
127.         txtProdi_3024.setText("");
128.         taVisualisasi_3024.setText("");
129.     }
130.
131.     private void btnDeleteActionPerformed_3024() {
132.         int selectedRow_3024 = tableMahasiswa_3024.getSelectedRow();
133.         if (selectedRow_3024 == -1) {
134.             JOptionPane.showMessageDialog(this, "Pilih baris yang
akan dihapus!", "Peringatan", JOptionPane.WARNING_MESSAGE);
135.             return;
136.         }
137.
138.         dataList_3024.remove(selectedRow_3024);
139.         refreshTable_3024();
140.         taVisualisasi_3024.setText("");

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141.     }
142.
143.     private void btnSortActionPerformed_3024() {
144.         if (dataList_3024.isEmpty()) {
145.             JOptionPane.showMessageDialog(this, "Tidak ada data
146.             untuk diurutkan!", "Info", JOptionPane.INFORMATION_MESSAGE);
147.             return;
148.         }
149.
150.         String algo_3024 = (String) cmbAlgo_3024.getSelectedItem();
151.         ArrayList<String> steps_3024 = new ArrayList<>();
152.
153.         ArrayList<Mahasiswa_2511533024> workingList_3024 = new
154.         ArrayList<>(dataList_3024);
155.
156.         steps_3024.add("Data Awal : " +
157.             formatList_3024(workingList_3024));
158.         steps_3024.add("");
159.
160.         if ("Insertion Sort".equals(algo_3024)) {
161.             steps_3024.add("=== INSERTION SORT ===");
162.             insertionSort_3024(workingList_3024, steps_3024);
163.         } else if ("Selection Sort".equals(algo_3024)) {
164.             steps_3024.add("=== SELECTION SORT ===");
165.             selectionSort_3024(workingList_3024, steps_3024);
166.         } else if ("Bubble Sort".equals(algo_3024)) {
167.             steps_3024.add("=== BUBBLE SORT ===");
168.             bubbleSort_3024(workingList_3024, steps_3024);
169.         }
170.
171.         steps_3024.add("");
172.         steps_3024.add("Hasil Akhir : " +
173.             formatList_3024(workingList_3024));
174.
175.         taVisualisasi_3024.setText(String.join("\n", steps_3024));
176.
177.         dataList_3024.clear();
178.         dataList_3024.addAll(workingList_3024);
179.         refreshTable_3024();
180.     }
181.
182.     private void insertionSort_3024(ArrayList<Mahasiswa_2511533024>
183.     list_3024, ArrayList<String> steps_3024) {
184.         int n_3024 = list_3024.size();
185.
186.         for (int i_3024 = 1; i_3024 < n_3024; i_3024++) {
187.             Mahasiswa_2511533024 key_3024 = list_3024.get(i_3024);
188.             int j_3024 = i_3024 - 1;
189.
190.             steps_3024.add("Key = " + key_3024.toString());
191.
192.             while (j_3024 >= 0 &&

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188.         key_3024.getName_3024().compareToIgnoreCase(list_3024.get(j_3024).getName_3
189.             list_3024.set(j_3024 + 1, list_3024.get(j_3024));
190.             j_3024--;
191.         }
192.
193.         list_3024.set(j_3024 + 1, key_3024);
194.         steps_3024.add("Langkah " + i_3024 + " : " +
formatList_3024(list_3024));
195.         steps_3024.add("");
196.     }
197. }
198.
199.     private void selectionSort_3024(ArrayList<Mahasiswa_2511533024>
list_3024, ArrayList<String> steps_3024) {
200.         int n_3024 = list_3024.size();
201.
202.         for (int i_3024 = 0; i_3024 < n_3024 - 1; i_3024++) {
203.             int minIdx_3024 = i_3024;
204.
205.             for (int j_3024 = i_3024 + 1; j_3024 < n_3024; j_3024++)
206.             {
207.                 if (list_3024.get(j_3024).getName_3024()
208.                     .compareToIgnoreCase(list_3024.get(minIdx_3024).getName_3024()) < 0) {
209.                     minIdx_3024 = j_3024;
210.                 }
211.
212.                 if (minIdx_3024 != i_3024) {
213.                     Collections.swap(list_3024, i_3024, minIdx_3024);
214.                     steps_3024.add("Swap index " + i_3024 + " dan " +
minIdx_3024);
215.                 } else {
216.                     steps_3024.add("Tidak ada swap pada pass ini");
217.                 }
218.
219.                 steps_3024.add("Pass " + (i_3024 + 1) + " : " +
formatList_3024(list_3024));
220.                 steps_3024.add("");
221.             }
222.         }
223.
224.     private void bubbleSort_3024(ArrayList<Mahasiswa_2511533024>
list_3024, ArrayList<String> steps_3024) {
225.         int n_3024 = list_3024.size();
226.
227.         for (int i_3024 = 0; i_3024 < n_3024 - 1; i_3024++) {
228.             boolean swapped_3024 = false;
229.
230.             for (int j_3024 = 0; j_3024 < n_3024 - i_3024 - 1;
j_3024++) {
231.                 if (list_3024.get(j_3024).getName_3024()

```

```

232.                                     .compareToIgnoreCase(list_3024.get(j_3024 +
233. 1).getName_3024()) > 0) {
234.                                     Collections.swap(list_3024, j_3024, j_3024 + 1);
235.                                     swapped_3024 = true;
236.
237.                                     steps_3024.add("Swap: " +
238. list_3024.get(j_3024).toString()
239.                                     + " <-> " + list_3024.get(j_3024 +
240. 1).toString());
241.                                     }
242.                                     }
243.                                     steps_3024.add("Pass " + (i_3024 + 1) + " : " +
244. formatList_3024(list_3024));
245.                                     steps_3024.add("");
246.                                     if (!swapped_3024) {
247.                                         steps_3024.add("Array sudah terurut, bubble sort
248. berhenti lebih cepat.");
249.                                         break;
250.                                     }
251.                                     }
252.                                     }
253.                                     private String formatList_3024(ArrayList<Mahasiswa_2511533024>
254. list_3024) {
255.                                         StringBuilder sb_3024 = new StringBuilder("[");
256.                                         for (int i_3024 = 0; i_3024 < list_3024.size(); i_3024++) {
257.                                             sb_3024.append(list_3024.get(i_3024).toString());
258.                                             if (i_3024 < list_3024.size() - 1) sb_3024.append(", ");
259.                                         }
260.                                         sb_3024.append("]");
261.                                         return sb_3024.toString();
262.                                     }
263.                                     public static void main(String[] args_3024) {
264.                                         SwingUtilities.invokeLater(() -> new
265. SortingProgram_2511533024().setVisible(true));
266.                                     }

```


3. HASIL OUTPUT

Sorting Mahasiswa - 2511533024

Input Data Mahasiswa

Nama:

NIM:

Program Studi:

Tambah

Pengurutan

Pilih Algoritma: **Insertion Sort** **Mulai Sorting** **Hapus Terpilih**

Daftar Mahasiswa

Nama	NIM	Program Studi
ain	008	kedokteran
andi	004	Teknik Elektro
Budi	001	Informatika
Citra	005	Manajemen Informatika
Maya	003	Ilmu Komputer
Rifal	024	informatika
rizky	002	Sistem Informasi

Proses Sorting (Langkah-langkah)

Key = Rifal (024)
Langkah 5 : [andi(004), Budi(001), Citra(005), Maya(003), Rifal(024), rizky(002), ain(008)]

Key = ain(008)
Langkah 6 : [ain(008), andi(004), Budi(001), Citra(005), Maya(003), Rifal(024), rizky(002)]

Hasil Akhir : [ain(008), andi(004), Budi(001), Citra(005), Maya(003), Rifal(024), rizky(002)]

Sorting Mahasiswa - 2511533024

Input Data Mahasiswa

Nama:

NIM:

Program Studi:

Tambah

Pengurutan

Pilih Algoritma: **Selection Sort** **Mulai Sorting** **Hapus Terpilih**

Daftar Mahasiswa

Nama	NIM	Program Studi
ain	008	kedokteran
andi	004	Teknik Elektro
Budi	001	Informatika
Citra	005	Manajemen Informatika
Maya	003	Ilmu Komputer
Rifal	024	informatika
rizky	002	Sistem Informasi

Proses Sorting (Langkah-langkah)

Tidak ada swap pada pass ini
Pass 5 : [ain(008), andi(004), Budi(001), Citra(005), Maya(003), Rifal(024), rizky(002)]

Tidak ada swap pada pass ini
Pass 6 : [ain(008), andi(004), Budi(001), Citra(005), Maya(003), Rifal(024), rizky(002)]

Hasil Akhir : [ain(008), andi(004), Budi(001), Citra(005), Maya(003), Rifal(024), rizky(002)]

Input Data Mahasiswa

Nama:
NIM:
Program Studi:

Pengurutan

Pilih Algoritma:

Daftar Mahasiswa

Nama	NIM	Program Studi
ain	008	kedokteran
andi	004	Teknik Elektro
Budi	001	Informatika
Citra	005	Manajemen Informatika
Maya	003	Ilmu Komputer
Rifal	024	informatika
rizky	002	Sistem Informasi

Proses Sorting (Langkah-langkah)

Data Awal : [ain(008), andi(004), Budi(001), Citra(005), Maya(003), Rifal(024), rizky(002)]

=== BUBBLE SORT ===
Pass 1 : [ain(008), andi(004), Budi(001), Citra(005), Maya(003), Rifal(024), rizky(002)]

Array sudah terurut, bubble sort berhenti lebih cepat.

Hasil Akhir : [ain(008), andi(004), Budi(001), Citra(005), Maya(003), Rifal(024), rizky(002)]

6. PENJELASAN SINGKAT PROGRAM

Program yang dibuat merupakan aplikasi berbasis GUI (Java Swing) untuk mengelola dan mengurutkan data mahasiswa yang terdiri dari nama, NIM, dan program studi. Program menyediakan fitur untuk menambahkan data mahasiswa baru, menghapus data yang dipilih pada tabel, serta melakukan proses sorting menggunakan tiga algoritma yaitu Insertion Sort, Selection Sort, dan Bubble Sort. Data mahasiswa ditampilkan dalam bentuk tabel (JTable), sedangkan proses pengurutan ditampilkan dalam area teks (JTextArea) sehingga pengguna dapat melihat langkah-langkah sorting secara bertahap. Proses sorting dilakukan berdasarkan nama mahasiswa dengan metode perbandingan `compareToIgnoreCase()`, sehingga pengurutan tidak dipengaruhi huruf besar atau kecil. Setelah sorting selesai, hasil akhir ditampilkan kembali pada tabel sehingga data mahasiswa menjadi tersusun rapi secara alfabetis sesuai algoritma yang dipilih.